

Homework 6

You may submit your solution to this homework assignment in the text window in iCollege, as a `.txt` file, or in some other commonly used format, *e.g.*, `.pdf`, `.docx`, *etc.*

1. Consider the language composed of all strings over the alphabet $\{a, b\}$ of the form: zero or more a 's followed by zero or more b 's, *i.e.*, $\{\varepsilon, a, b, aa, ab, bb, aaa, aab, abb, bbb, \dots\}$, where ε is the empty string. This language is a regular language: show why this is. Note that you may use any line of argument from the slides on regular languages, *e.g.*, by
 - (a) showing that it can be constructed using the properties on slide 5 (*Regular Language: a formal definition*), like in the examples on the subsequent slides 6 and 7, or by
 - (b) drawing the finite state automaton (FSA), or giving its description, like in the examples on slides 13 and 14. If you draw the FSA, you may draw in on a piece of paper, take a picture, and upload that as part of the assignment.
2. Consider the language composed of all strings over the alphabet $\{a, b\}$ that contain an *odd number* of a 's. Is this a regular language? If so, show why. Note, again, that you may use any line of argument from the slides on regular languages, *e.g.*, by
 - (a) using the properties on slide 5, by
 - (b) drawing the FSA, or by
 - (c) using the additional properties on slide 9 (*Closure properties of regular languages*), along with the fact that the language on slide 7 is regular (*hint, hint, ...*).
3.
 - (a) What would be a regular expression (regex) for the language of exercise 1.?
 - (b) What would be a regex for the language of exercise 2.?
 - (c) Can you think of another (different, but equivalent) regex for (b) above?
4. Using your favorite implementation of regular expressions (Python, grep, sed, ...), input your regex from exercise 3(a), and one of your regexes from exercise 3(b) (or 3(c), since they are equivalent they should generate the same result anyway), and test to see how many strings from the set of all strings over the alphabet $\{a, b\}$ of length at most 5, *i.e.*, $\{\varepsilon, a, b, ab, \dots, bbbbb\}$ are described by, that is, match, each of your input regexes above. *Note:* that you will have to match the *entire* string with the regex, so watch out for this implementation detail. How many strings (you can just give the count) are matched with
 - (a) your regex from exercise 3(a)?
 - (b) your regex from exercise 3(b) (or 3(c))?

Note: that you may use <https://github.com/murraypatterson/regex-example> for this purpose: it already generates the set of (63) strings mentioned above, and uses the regex `b*(ab*ab*)*` for all strings over alphabet $\{a, b\}$ with an even number of a 's (you can just replace this with your regex). Moreover, it guarantees a match of the *entire* string (by using Python's `fullmatch`), and so it is a safe bet to use for this assignment: just plug in your regex, and see how many strings match.

Also: note that 32 of the 63 strings (including the empty string) are matched with the current `b*(ab*ab*)*` regex. However many of these 63 should match in a regex for exercises 3(b) (or 3(c)) would be the answer to (a) and (b) above.